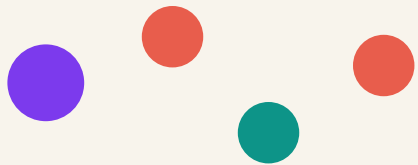


# Nearest Neighbors

*K-NN · Distance-Weighted NN · Theory & Boundaries*



Non-parametric · instance-based · no training phase

# Arun Chauhan, CSE, GEU, ML

# What is a non-parametric model?

**A model whose complexity grows with the data.  
It does not assume a fixed functional form (no fixed number of parameters).**

## Parametric

- Fixed number of parameters
- Assumes a form (linear, logistic...)
- Fast prediction after training
- Examples: linear regression, logistic regression, Naïve Bayes

## Non-parametric

- “Parameters” = the data itself
- No fixed functional form
- Prediction needs the whole set
- Examples: K-NN, decision trees, kernel density, Gaussian processes

# The classification rule

Given query  $x$ , let  $N_k(x)$  be the  $k$  nearest training points.

$$\hat{y} = \operatorname{argmax}_c \sum_{i \in N_k(x)} \mathbf{1}(y_i = c)$$

Majority class among the  $k$  nearest neighbors.

**1**

## Compute distance

$d(x, x_i)$  for every  
training point  $x_i$

**2**

## Select neighbors

Keep the  $k$  points  
with smallest  $d$

**3**

## Majority vote

Assign the class that  
appears most often

## KNN is a pure memory-based classifier

### No training

“Training” is just storing the data.  
No weights, no loss.

### Model = data

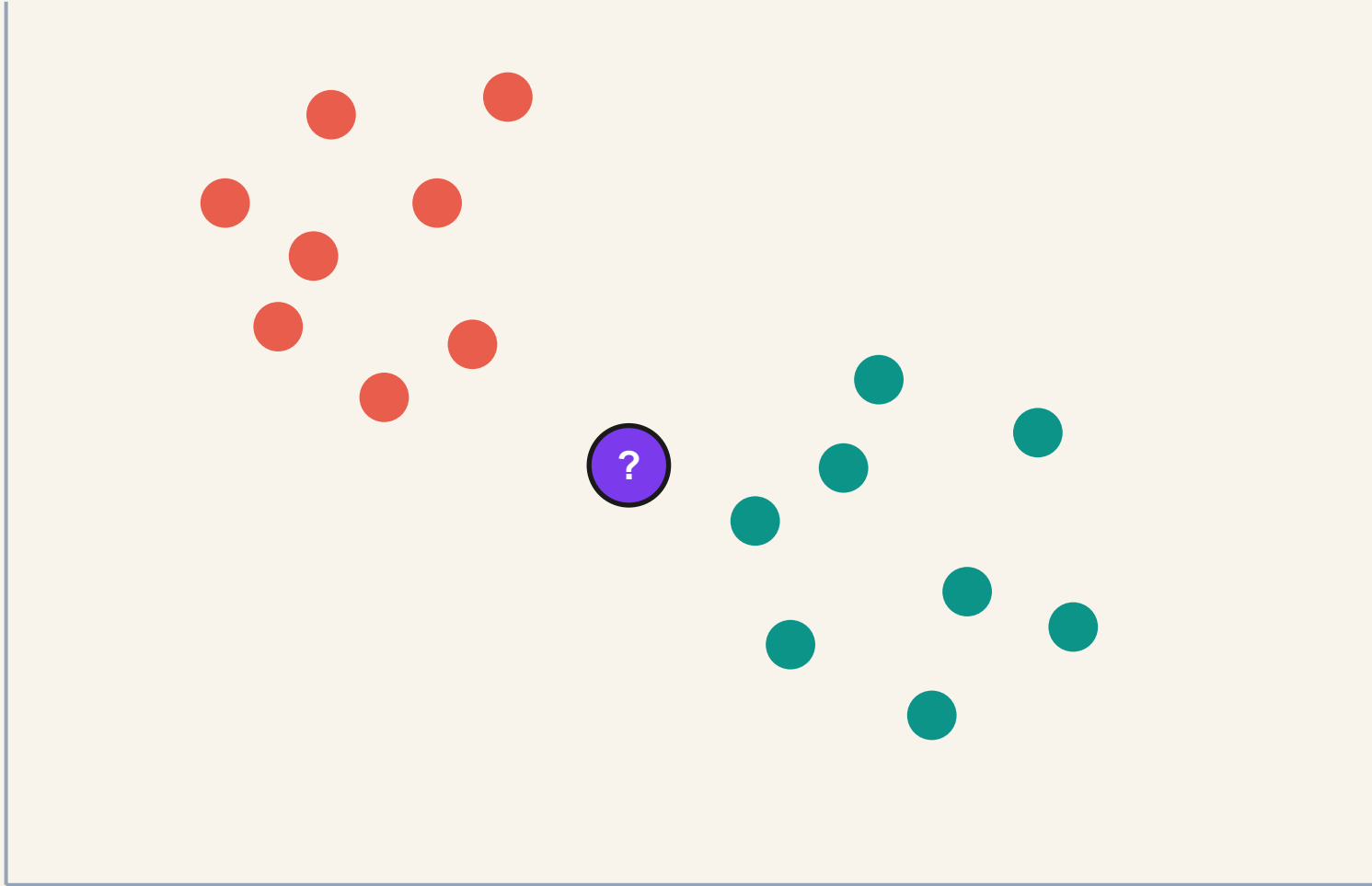
The decision surface is defined only when a query arrives.

### Complexity ↑

More points → more complex boundary.  
Capacity grows with  $n$ .

Consequence: prediction cost is  $O(n \cdot d)$  for naïve search.  
That is the price of being non-parametric and lazy.

## Look at the neighbors of a new point

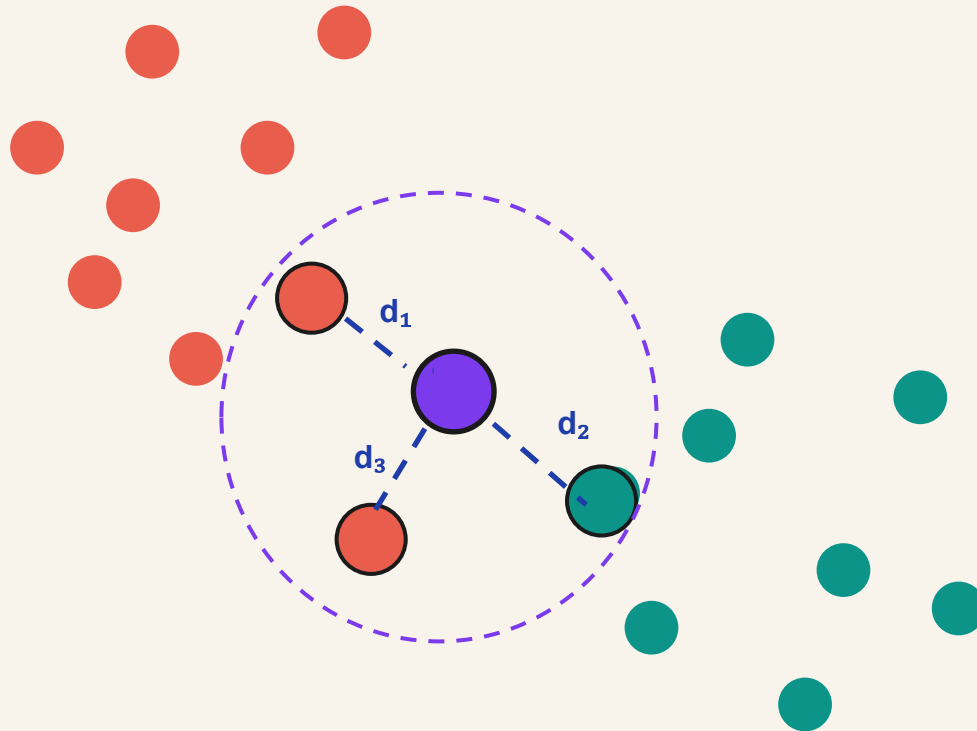


**What class is ?**

-  Class 1
-  Class 0
-  Query point

Find the closest training points and let them vote.

**k = 3** · connect to the closest three



**3 nearest**

● 1st · Class 1

● 2nd · Class 0

● 3rd · Class 1

**Majority: Class 1**  
(2 out of 3)

Each of the k neighbors gets one equal vote

$$\hat{y} = \text{mode} \{ y_1, y_2, \dots, y_k \}$$

Neighbor 1

**Class 1**

1 vote

Neighbor 2

**Class 0**

1 vote

Neighbor 3

**Class 1**

1 vote

**Result: Class 1 (2 votes > 1 vote)**

## 1-NN = Voronoi diagram of the training set



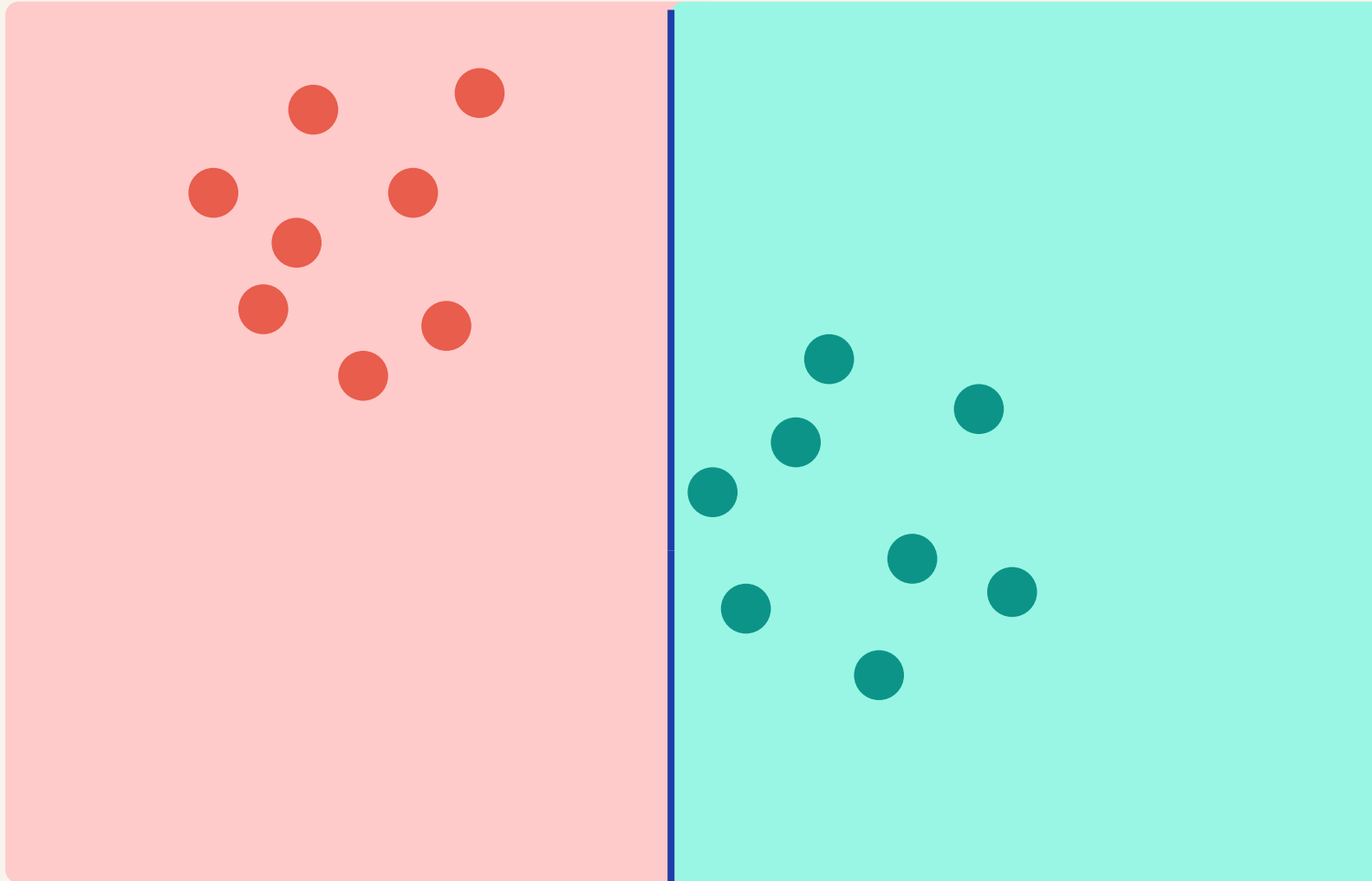
**$k = 1$**

Every location is assigned the label of its single nearest training point.

The boundary is the set of points equidistant to two training points of different classes (perpendicular bisectors).

- Maximum flexibility
- High variance
- Sensitive to outliers

## Smoother — majority over a local neighborhood



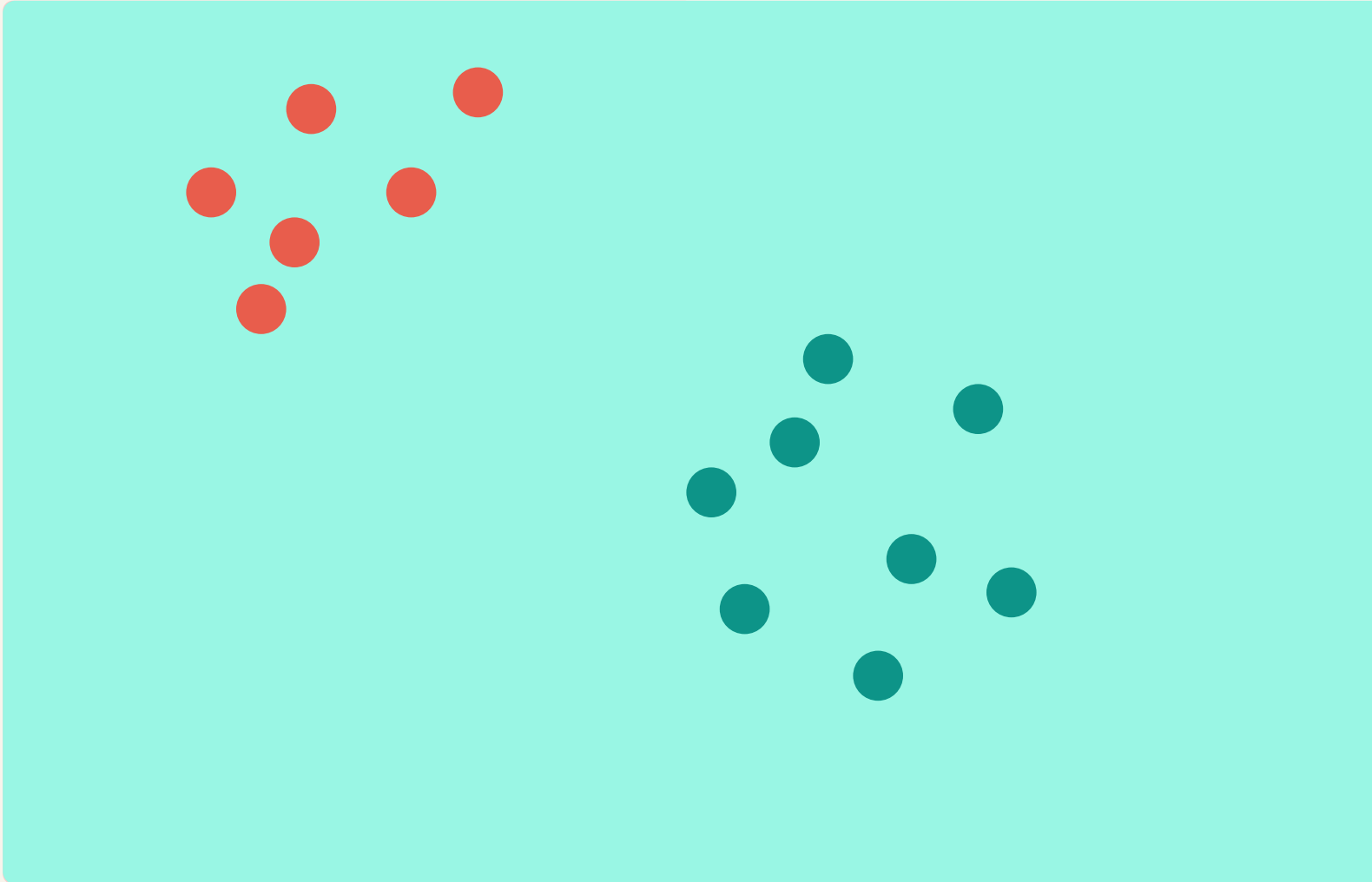
**$k = 5$**

Each location looks at its 5 nearest points and takes a majority vote.

Isolated points lose their private cells. The boundary becomes continuous and less noisy.

- Balanced bias-variance
- Common practical choice

## Almost a global half-plane — high bias



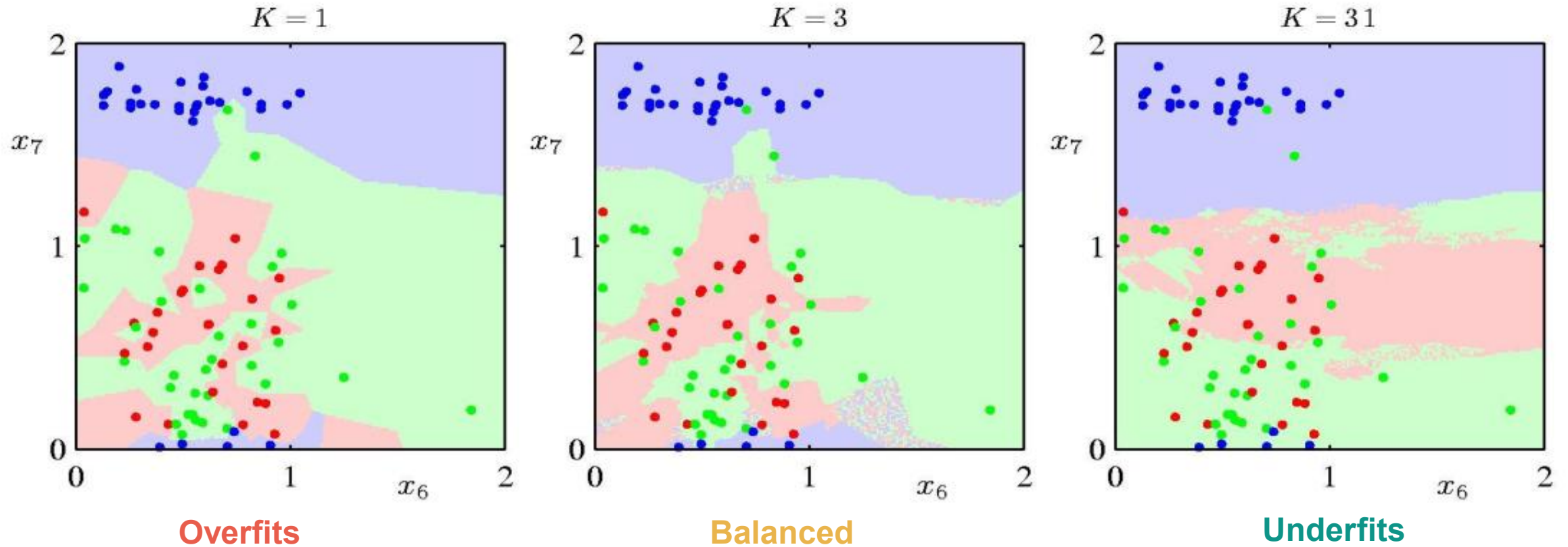
**$k = N$**

When  $k$  approaches  $n$ , every query sees almost the whole training set.

The decision reduces to the global majority class (or a very smooth separator).

- Very low variance
- High bias
- Can underfit

## How the decision surface changes with $k$



- $K$  acts as a smoother

# How we measure “near”

## Euclidean

Straight-line distance

$$d = \sqrt{\sum (x_j - y_j)^2}$$

Most common default.  
Sensitive to scale —  
normalize features first.

## Manhattan

Grid / city-block distance

$$d = \sum |x_j - y_j|$$

More robust to outliers  
in high dimensions.  
Useful when axes matter.

## Closer neighbors get higher weight

$$w_i = 1 / d_i \quad (\text{or } 1 / d_i^2)$$

Closer  $\rightarrow$  larger weight

Score for class  $c$

$$S(c) = \sum w_i \cdot \mathbb{1}(y_i = c) \quad \rightarrow \quad \hat{y} = \operatorname{argmax}_c S(c)$$

### Neighbor A

Class 1  
 $d = 0.5 \rightarrow w = 2.0$

### Neighbor B

Class 0  
 $d = 1.0 \rightarrow w = 1.0$

### Neighbor C

Class 1  
 $d = 2.0 \rightarrow w = 0.5$

# KNN in one page

**Non-parametric**

Model complexity grows with  $n$ . No fixed parameter vector.

**Lazy / instance-based**

All computation happens at query time. Training = store data.

**Classification rule**

$\hat{y}$  = majority (or weighted majority) of the  $k$  nearest labels.

 **$k$  controls smoothness**

$k=1 \rightarrow$  Voronoi (high variance). Large  $k \rightarrow$  smooth (high bias).

**Distance weighting**

$w_i = 1/d_i$  lets close points dominate without discarding far ones.

**Just for research and project; not for exam.**

RECENT RESEARCH

# **A\* & arXiv Papers on kNN**

*NeurIPS · ICML · ICLR · Nature Comm. · arXiv (2025–2026)*

**Consistency · ModernNCA · Private kNN · kNN Attention****NeurIPS 2025****Adaptive sampling consistency**

$k_n$ -NN is online consistent under mild smoothness even with adaptive queries.

**ICLR 2025****ModernNCA (tabular)**

Soft differentiable kNN + deep embeddings matches CatBoost on 300 datasets.

**ICML 2025****Kona: private kNN**

Secure multi-party kNN; up to 3000× less communication.

**ICLR 2025****kNN Attention**

Theory for sparse attention: each token attends to  $k$  nearest tokens.

## Boltzmann margin · Gromov–Wasserstein kNN

arXiv 2606.10361

### Near-Exponential Rates via Boltzmann Margin

Yang, Shafaei, Lan

New margin condition between Tsybakov and Massart.

First near-exponential convergence rates for kNN classification — stronger than classical polynomial rates under weaker assumptions.

arXiv 2606.10295

### kNN in Gromov–Wasserstein Space

Hohmeier, Fraiman, Moosmueller

kNN with GW / fGW distance on metric-measure spaces and graphs.

Proves universal consistency for graphs and node-attributed graphs.

## Adaptive graphs · Manifold search · Coresets

**Nature Comm. 2026**

**Adaptive kNN graph**

HNSW + pre-computed adaptive voting. Real-time inference without accuracy loss (Li, Xu, Zhang).

**arXiv 2605.02224**

**Manifold k-NN**

Extends DP nearest-neighbor search to arbitrary  $k$  on manifold data. 1–10× faster than kd-trees.

**arXiv 2608.16270**

**Coreset via kNN graphs**

Efficient coreset selection using kNN graphs for data summarization / training-set reduction.

## LLM routing · Adaptive / robust classifiers

ICLR 2026 (submitted)

### Simple kNN Beats Learned LLM Routers

Yang Li

Well-tuned kNN router matches or beats complex learned routers for choosing which LLM to call.

Locality in embedding space → non-parametric methods need fewer samples. New multi-modal routing benchmarks.

arXiv / Soft Computing 2026

### Adaptive & robust kNN variants

- Granular-ball adaptive-k classifier (arXiv 2608.12903)
- IWUkNN: iterative weight updates against outliers & imbalance (Appl. Soft Comp. 2026)

Strong gains on dozens of datasets.

# kNN research at a glance

|              |                            |                                                                       |
|--------------|----------------------------|-----------------------------------------------------------------------|
| 2025         | Consistency & deep kNN     | Adaptive sampling; ModernNCA; private kNN; kNN attention              |
| 2026 theory  | Boltzmann margin · GW-kNN  | Near-exponential rates; universal consistency on graphs               |
| 2026 systems | Adaptive graphs · Manifold | Real-time inference; faster manifold queries; coresets                |
| 2026 apps    | LLM routing · Robust kNN   | Simple kNN beats learned routers; adaptive / imbalance-aware variants |

## KEY TAKEAWAY

# kNN is still evolving

Majority vote estimates local class probability (Bayes under 0-1 loss).  
Theory (margins, consistency), systems (graphs, manifolds), and applications (LLM routing, robust variants) all build on the same simple idea.

**Questions?**